Term Structure Modeling and Forecasting Using the Nelson-Siegel Model

39

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Abstract

In this chapter, we illustrate some recent developments in the yield curve modeling by introducing a latent factor model called the dynamic Nelson-Siegel model. This model not only provides good in-sample fit, but also produces superior out-of-sample performance. Beyond Treasury yield curve, the model can also be useful for other assets such as corporate bond and volatility. Moreover, the model also suggests generalized duration components corresponding to the level, slope, and curvature risk factors.

The dynamic Nelson-Siegel model can be estimated via a one-step procedure, like the Kalman filter, which can also easily accommodate other variables of interests. Alternatively, we could estimate the model through a two-step process by fixing one parameter and estimating with ordinary least squares. The model is flexible and capable of replicating a variety of yield curve shapes: upward

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sloping, downward sloping, humped, and inverted humped. Forecasting the yield curve is achieved through forecasting the factors and we can impose either a univariate autoregressive structure or a vector autoregressive structure on the factors.

Keywords
Term structure • Yield curve • Factor model • Nelson-Siegel curve • State-space model

39.1 Introduction

There have been major advances in theoretical term structure models, as well as their econometric estimation. Two popular approaches to term structure modeling are no-arbitrage models and equilibrium models. The no-arbitrage tradition focuses on eliminating arbitrage opportunities by perfectly fitting the term structure at a point in time, which is important for pricing derivatives; see Hull and White (1990) and Health et al. (1992), among others. The equilibrium tradition models the dynamics of the instantaneous rate typically through affine models so that yields at other maturities can be derived under various assumptions about the risk premium (e.g., Vasicek 1977; Cox et al. 1985; Duffie and Kan 1996).

For many finance questions, such as bond portfolio management, derivatives pricing, and risk management, it is crucial to both produce accurate estimates of the current term structure as well as forecast future interest rate dynamics. One class of models that has one potential satisfactory answer to these questions is that of the Nelson-Siegel class of models (see, Nelson and Siegel 1987). Here, we survey some recent developments of this model. This model not only provides good in-sample fit, but also produces superior out-of-sample performance. Moreover, the model also suggests generalized duration components corresponding to the level, slope, and curvature risk factors.

39.2 Modeling the Term Structure

Let $P_t(\tau)$ be the date t price of a zero-coupon riskless bond that pays \$1 in τ periods. Then,

$$P_t(\tau) = \exp(-\tau y_t(\tau)). \tag{39.1}$$

In practice, yield curves, discount curves, and forward curves are not observed. Instead, they must be estimated from observed bond prices. A popular approach is to estimate forward rates at the observed maturities, and then, construct unsmoothed yields by averaging appropriate estimated forward rates. These yields exactly price the included bonds (see Fama and Bliss 1987).

The original Nelson and Siegel (1987) framework is a convenient and parsimonious three-component exponential approximation. They work with the forward

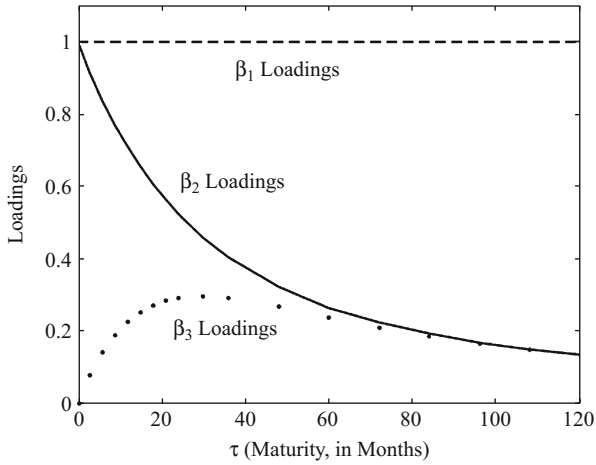

Fig. 39.1 Factor loadings. This figure plots the factor loadings when λ is prefixed at 0.0609

rate, which can be viewed as a constant plus a Laguerre function. Diebold and Li (2006) made it dynamic, and thus, the corresponding yield curve is

$$y_t(\tau) = \beta_{1t} + \beta_{2t}\left(\frac{1 - e^{-\lambda\tau}}{\lambda\tau}\right) + \beta_{3t}\left(\frac{1 - e^{-\lambda\tau}}{\lambda\tau} - e^{-\lambda\tau}\right) + \varepsilon_t, \tag{39.2}$$

and call it DNS hereafter. β_1 changes all yields uniformly, and it can be called the level factor (L_t). β_2 loads the short rate more heavily, its loading decays to zero as maturity lengthens, and it can be called the slope factor (S_t). β_3 loads the medium term more heavily, its loading starts at zero and decays back to zero as maturity increases, and it can be called the curvature factor (C_t). In Fig. 39.1, we plot the factor loadings for a specific value of λ. λ determines the maturity at which the medium-term (or the curvature factor) loading achieves its maximum, which effectively controls the location of the hump of the curve. The DNS model is different from the factor analysis, in which one estimates both the unobserved factors and factor loadings. Here, the framework imposes structure on the factor loadings.

39.3 Fitting the Yield Curve

39.3.1 Data Construction

The standard way of measuring the term structure is by means of the spot rate curve (or equivalently the yield-to-maturity) on zero-coupon bonds.[1] The problem with

[1]Zero coupon bonds are chosen to limit the "coupon effect," which implies that two bonds that are identical in every respect except for bearing different coupon rates can have a different yield-to-maturity.

zero-coupon yields is that these are not usually directly observed. For example, the US Treasury Bills do not bear coupon, but they are only available for maturities of 1-year or less, and the longer maturity zero-coupon yields need to derived from coupon bearing Treasury Notes and Bonds. A popular approach to resolve such an issue is the bootstrapping procedure by Fama and Bliss (1987), which sequentially extracts forward rates from bond prices with successively longer maturities and then takes advantages of the interchangeable nature of the spot rate curve, discount curve, and forward rate curve.

39.3.2 Estimation Procedure

Because of the structure the model imposes, estimation can be achieved with high precision. There are two popular approaches: the one-step and two-step procedures. The one-step procedure can be achieved in two ways. One method is simply to estimate the model by nonlinear least squares for each month t. The other way is to transform the system into a state-space representation and estimate λ and the factors via a Kalman filter. Alternatively, for a two-step procedure, Diebold and Li (2006) advocate prefixing the λ, and estimating the factors via ordinary least squares (OLS). The appendix provides more discussion of these two approaches.

As we can see in Fig. 39.2, the model is flexible and is capable of replicating a variety of yield curve shapes: upward sloping, downward sloping, humped, and inverted humped. Figure 39.3 displays the autocorrelations of the estimated factors and residuals. The level factor displays high persistence and is, of course, positive. In contrast, the slope and curvature factors are less persistent and assume both positive and negative values.

39.4 Forecasting the Yield Curve

Forecasting the yield curve can be achieved through forecasting the factors. We can impose either a univariate autoregressive structure or a vector autoregressive structure on the factors.

The yield forecasts based on underlying univariate AR(1) factor specifications are

$$\hat{y}_{t+h/t}(\tau) = \beta_{1,t+h/t} + \beta_{2,t+h/t}\left(\frac{1-e^{-\lambda_t\tau}}{\lambda_t\tau}\right) + \beta_{3,t+h/t}\left(\frac{1-e^{-\lambda_t\tau}}{\lambda_t\tau} - e^{-\lambda_t\tau}\right), \quad (39.3)$$

where β forecasts are made by the following:

$$\begin{aligned} \hat{\beta}_{1,t+h/t} &= \hat{c}_1 + \hat{\gamma}_1\hat{\beta}_{1t} \\ \hat{\beta}_{2,t+h/t} &= \hat{c}_2 + \hat{\gamma}_2\hat{\beta}_{2t} \\ \hat{\beta}_{3,t+h/t} &= \hat{c}_3 + \hat{\gamma}_3\hat{\beta}_{3t} \end{aligned} \quad (39.4)$$

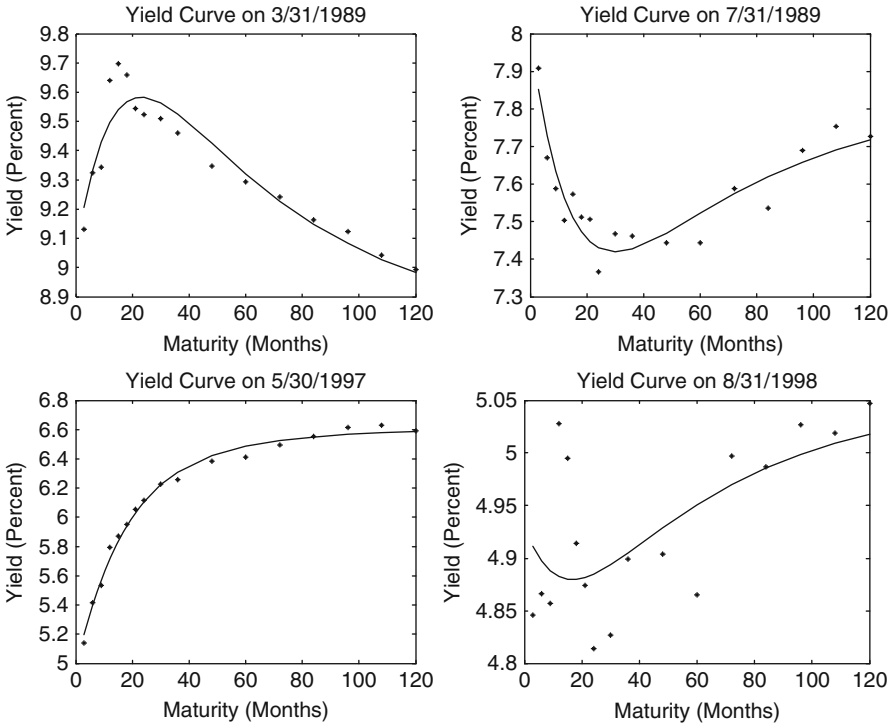

Fig. 39.2 Selected fitted yield curves. This figure presents fitted yield curve for selected dates, together with actual yields (Source: Diebold and Li 2006)

The yield forecasts based on an underlying multivariate autoregressive specification (Vector AR(1)) are

$$\hat{y}_{t+h/t}(\tau) = \beta_{1,t+h/t} + \beta_{2,t+h/t}\left(\frac{1-e^{-\lambda_t\tau}}{\lambda_t\tau}\right) + \beta_{3,t+h/t}\left(\frac{1-e^{-\lambda_t\tau}}{\lambda_t\tau} - e^{-\lambda_t\tau}\right), \quad (39.5)$$

where

$$\hat{\beta}_{t+h/t} = \hat{c} + \hat{\gamma}\hat{\beta}_t, \quad (39.6)$$

and $\beta = \{\beta_1, \beta_2, \beta_{3,}\}'$.

The random walk model,

$$\hat{y}_{t+h/t}(\tau) = y_t(\tau), \quad (39.7)$$

is a no-change forecast model, which is used as the benchmark.[2]

[2]Diebold and Li (2006) conduct a more extensive comparisons with competing models.

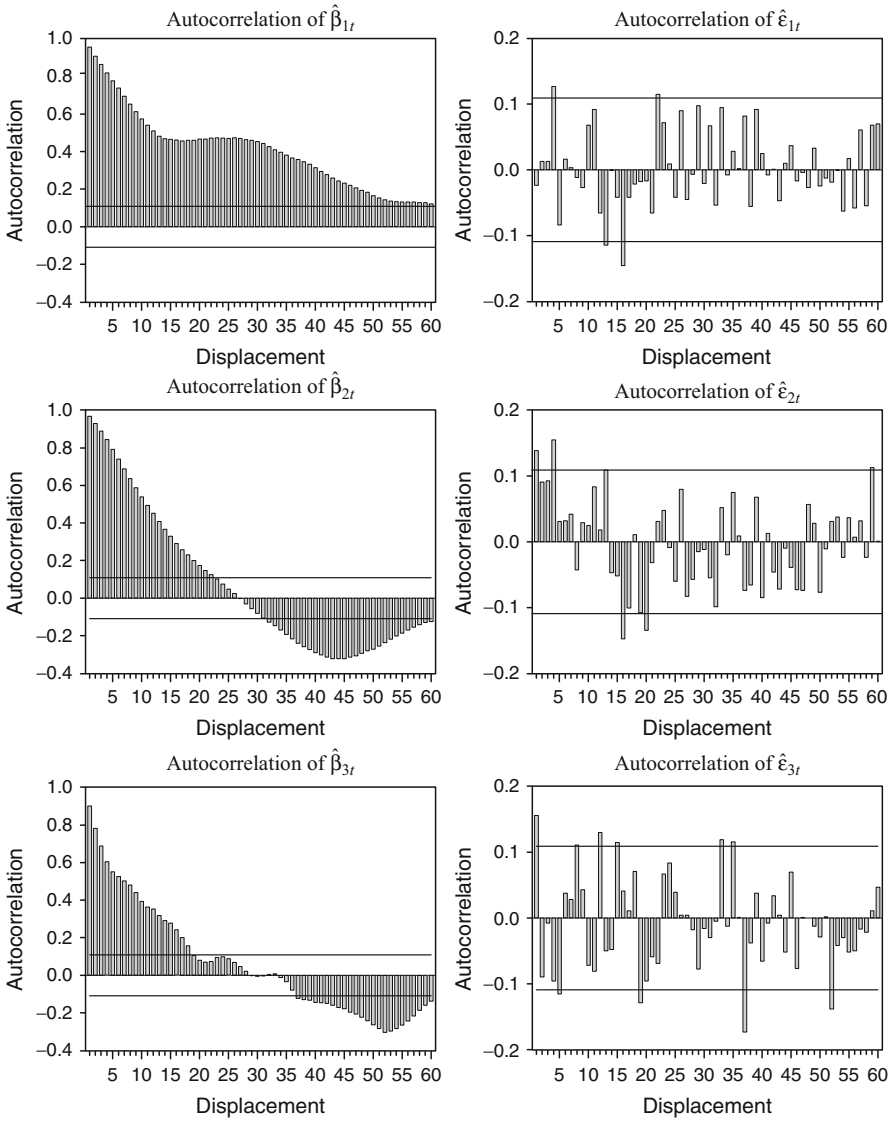

Fig. 39.3 Autocorrelation and residual autocorrelation. This figure presents sample autocorrelations of the level, slope, and curvature factors, as well as the sample autocorrelations of AR (1) models t to the three estimated factors, along with Barletts approximate 95 condence bands (Source: Diebold and Li 2006)

We define forecast errors at $t + h$ as $y_{t+h}(\tau) - \hat{y}_{t+h|t}(\tau)$. Table 39.1 reports the root mean-squared errors of the out-of-sample performance of the DNS model versus the random walk model. An error of 0.235 indicates a mean-squared error of 23.5 basis points in the yield prediction. For 1 month ahead, the random walk is

Table 39.1 Out-of-sample forecasting results

	3-month	1-year	2-year	3-year	5-year	10-year	30-year
1 *month a head*							
Random walk	0.235	**0.259**	**0.290**	0.300	**0.288**	**0.257**	**0.211**
Unrestricted VAR	**0.213**	0.261	0.307	0.302	0.310	0.314	0.419
Univariate AR	0.248	0.282	0.315	0.301	0.311	0.314	0.418
6 *months ahead*							
Random walk	0.910	0.904	0.938	**0.857**	**0.813**	0.655	0.570
Unrestricted VAR	0.909	1.050	1.078	0.999	0.949	0.796	0.717
Univariate AR	**0.694**	**0.890**	**0.908**	0.881	0.827	**0.651**	**0.486**
1 years ahead							
Random walk	1.552	1.505	1.343	1.190	1.065	0.858	0.622
DNS Unrestricted VAR	1.643	1.731	1.641	1.483	1.338	1.123	0.979
DNS Univariate AR	**1.383**	**1.445**	**1.338**	**1.182**	**1.041**	**0.819**	**0.615**

This table presents the results of out-of-sample forecasts using VAR and univariate AR specifications of the DNS factors. I estimate all models recursively from 1987:1 to the time the forecast is made, beginning in 1997:1 and extending through 2002:12. I define forecast errors at $t + h$ as $y_{t+h}(\tau) - \hat{y}_{t+h/t}(\tau)$ and report the root mean-squared errors versus random walk.

hard to beat, but matters improve dramatically. The DNS model with an AR (1) specification produces smaller errors, so it clearly has advantages (see Diebold and Li 2006 for more details). Remember here, we only illustrate the point forecast, and it can be extended easily to interval forecasting since they are useful for risk management.

39.5 Other Applications

Beyond the US Treasury yield curve, the Nelson-Siegel model has also shown success in fitting and forecasting other assets and global yield curves (see Krishnan et al. 2010; Diebold et al. 2008; Hua 2010). Since the model produces three factors that capture the information in the entire term structure, we can analyze the dynamic relationship among them. For example, Hua (2010) analyzes the dynamic interaction between the credit spread term structure and equity option implied volatility term structure.

Moreover, the DNS framework can be easily augmented with other variables of interest. We could also analyze how macroeconomic variables are linked with the interest rate term structure; see Auroba et al. (2006). This can be done easily in a state-space framework. Frequently, yield variation reacts to macrovariables, and macrovariables can also be impacted by the yields. Potentially, we can improve our forecasts of one by incorporating information from the other.

39.6 Generalized Duration Measure

Traditional interest rate risk management focuses on duration and duration management, which only considers parallel shifts of the yield curve. However, in practice, nonparallel shifts do exist and these are a significant source of risk. The DNS model presents a generalized duration component that corresponds to the level, slope, and curvature risk factors.

We can define a bond duration measure as follows. Let the cash flows from bond be C_1,C_2, . . .,C_I, and define the associated maturities to be τ_1,τ_2,. . .,τ_I. We assume that the yield curve is linear in some arbitrary factors f_1, f_2, and f_3,

$$y_t(\tau) = B_1(\tau)f_{1t} + B_2(\tau)f_{2t} + B_3(\tau)f_{3t}. \tag{39.8}$$

This is consistent with the DNS setup. Since the price of the bond can be expressed as,

$$P = \sum_{i=1}^{I} C_i e^{-\tau_i y_t(\tau_i)}, \tag{39.9}$$

for an arbitrary change of the yield curve, the price change is

$$dP = \sum_{i=1}^{I} \frac{\partial P}{\partial y_t(\tau_i)} dy_i(\tau_i) = \sum_{i=1}^{I} \left[C_i e^{-\tau_i y_t(\tau_i)}(-\tau_i)\right] dy_t(\tau_i), \tag{39.10}$$

Where $y_t(\tau_i)$ are treated as independent variables. Rearranging terms, we can express the percentage change in bond price as a function of changes in the factors

$$-\frac{dP}{P} = \sum_{j=1}^{3} \left\{ \sum_{i=1}^{I} \left[\frac{1}{P} C_i e^{-\tau_i y_t(\tau_i)} \tau_i\right] B_j(\tau_i) \right\} df_{it} = \sum_{j=1}^{3} \left\{ \sum_{i=1}^{I} w_i \tau_i B_j(\tau_i) \right\} df_{it}, \tag{39.11}$$

where w_i is the weight associated with C_i.

Since we have decomposed the bond price changes into risk factor changes, the duration component associated with each risk factor is, for $j = 1,2,3$,

$$D_j = \sum_{i=1}^{I} w_i \tau_i B_j(\tau_i). \tag{39.12}$$

Moreover, based on the dynamic Nelson-Siegel model, the vector duration is

$$
\begin{aligned}
D_1 &= \sum_{i=1}^{I} w_i \tau_i \\
D_2 &= \sum_{i=1}^{I} w_i \frac{1 - e^{-\lambda \tau_i}}{\lambda} \\
D_3 &= \sum_{i=1}^{I} w_i \left(\frac{1 - e^{-\lambda \tau_i}}{\lambda} - \tau_i e^{-\lambda \tau_i} \right).
\end{aligned}
\tag{39.13}
$$

The vector duration measure has the following properties:

- D_1, D_2, and D_3 increase with maturity τ.
- D_1, D_2, and D_3 decrease with coupon rate.
- D_1, D_2, and D_3 decrease with yield-to-maturity.

Note that D_1 is exactly the tradition Macaulay duration.

Diebold et al. (2006) apply this vector duration measure as immunization tools in a practical bond portfolio management context. They report that hedging based on the vector duration outperforms hedging based on Macaulay duration in almost all samples and all holding periods. Moreover, it also outperforms polynomial vector duration during unusual market situations such as the monetary regime of the early 80s. Therefore, the vector duration measure seems to be an appealing risk management tool.

39.7 Summary

We have presented the DNS model as a capable and flexible model that can fit the term structure of interest rates in-sample and predict them out-of-sample. It can easily be extended to a different asset class or accommodate other variables into the specifications. Given the advantages of the dynamic Nelson-Siegel model in many dimensions, it is gaining in popularity. Recently, the Board of Governors of the Federal Reserve System has started publishing the daily estimated factors of the Treasury yield curve on their website.[3]

One potential drawback of the DNS model is that it does not rule out arbitrage opportunities. Recently, an arbitrage free version of the model has been developed; see Christensen et al. (2009, 2011).

Appendix 1: One-Step Estimation Method

We illustrate the one-step method to estimate the DNS model. If the dynamics of betas (factors) follow a vector autoregressive process of first order, the model

[3]The data can be found at http://www.federalreserve.gov/econresdata/researchdata.htm

immediately forms a state-space system since ARMA state vector dynamics of any order may be transformed into a state-space form. Thus, the transition equation is

$$\begin{pmatrix} \beta_{1,t} - \mu_{\beta 1} \\ \beta_{2,t} - \mu_{\beta 2} \\ \beta_{3,t} - \mu_{\beta 3} \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} \beta_{1,t-1} - \mu_{\beta 1} \\ \beta_{2,t-1} - \mu_{\beta 2} \\ \beta_{3,t-1} - \mu_{\beta 3} \end{pmatrix} + \begin{pmatrix} \eta_{1,t} \\ \eta_{2,t} \\ \eta_{3,t} \end{pmatrix}, \tag{39.14}$$

$t = 1, \ldots, T$. The measurement equation, which relates N yields to the three unobserved factors, is

$$\begin{pmatrix} y_t(\tau_1) \\ y_t(\tau_2) \\ \vdots \\ y_t(\tau_N) \end{pmatrix} = \begin{pmatrix} 1 & \dfrac{1 - e_1^{\tau}\lambda}{\tau_1 \lambda} & \dfrac{1 - e^{-\tau_1 \lambda}}{\tau_1 \lambda} - e^{-\tau_1 \lambda} \\ 1 & \dfrac{1 - e_2^{\tau}\lambda}{\tau_2 \lambda} & \dfrac{1 - e^{-\tau_2 \lambda}}{\tau_2 \lambda} - e^{-\tau_2 \lambda} \\ \vdots & \vdots & \vdots \\ 1 & \dfrac{1 - e_1^{\tau}\lambda}{\tau_N \lambda} & \dfrac{1 - e^{-\tau_N \lambda}}{\tau_N \lambda} - e^{-\tau_N \lambda} \end{pmatrix} \begin{pmatrix} \beta_{1,t} \\ \beta_{2,t} \\ \beta_{3,t} \end{pmatrix} + \begin{pmatrix} \varepsilon_{1,t} \\ \varepsilon_{2,t} \\ \vdots \\ \varepsilon_{\mathrm{N},t} \end{pmatrix}, \tag{39.15}$$

$t = 1, \ldots, T$. In matrix notation, we rewrite the state-space system as measurement equation,

$$y_t = \Gamma f_t + \varepsilon_t; \tag{39.16}$$

state equation,

$$(f_t - \mu) = A(f_{t-1} - \mu) + \eta_t. \tag{39.17}$$

For optimality of the Kalman filter, we assume that the white noise transaction and measurement errors be orthogonal to each other.

$$\begin{pmatrix} \eta_t \\ \varepsilon_t \end{pmatrix} \sim WN\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} Q & 0 \\ 0 & H \end{bmatrix} \right), \tag{39.18}$$

$$E\left(f_0 \eta_t^{'}\right) = 0, \tag{39.19}$$

$$E\left(f_0 \varepsilon_t^{'}\right) = 0. \tag{39.20}$$

The state-space set up with the application of the Kalman filter delivers maximum-likelihood estimates and smoothed underlying factors, where all parameters are estimated simultaneously. Such representation also allows for heteroskedasticity, missing data, or heavy-tailed measurement errors. Moreover, other useful variables, such as macroeconomic variables, can be augmented into the state equation to understand the dynamic interactions between the yield curve and the macroeconomy.

Appendix 2: Two-Step Estimation Method

The estimation for the two-step procedure requires us to choose a λ value first. Once the λ is fixed, the values of the two regressors (factor loadings) can be computed, so ordinary least squares can be applied to estimate the betas (factors) at each period t. Doing so is not only simple and convenient, but also eliminates the potential for numerical optimization challenges. The question is: What is the appropriate value of λ. Recall that λ determines the maturity at which the loading achieves its maximum. For example, for Treasury data, 2 or 3 years are commonly considered medium term, so a simple average of the two is 30-month. The λ value that maximizes the loadings on the medium-term factor at exactly 30-month is 0.0609. For a different market, the medium maturity could be different, so the corresponding λ value could be different as well.

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